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Research on clearance flow characteristics of gas-lubricated journal bearings using the lattice Boltzmann method

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Abstract Gas-lubricated journal bearings (GLJBs) are widely utilized in rotating machines and the complex flow in ultra-thin clearances is a hot research topic. In this paper, the micro-clearance between the journal and housing of a GLJB was selected as the research object of a typically curved variable cross-section microchannel. In addition, the multiple-relaxation-time (MRT) model of the lattice Boltzmann method (LBM) was employed to analyze the shear flow characteristics. The generalized form of interpolation supplemented LBM (GILBM) based on the body-fitted grid was utilized due to complex geometry, and the Langmuir slip boundary model was introduced through the non-equilibrium extrapolation (NEE) method to simulate the rarefied airflow from the slip region to transition region. The velocity and pressure fields in the microchannel under different bearing parameters and operating conditions were described by the present model. What's more, the gas backflow phenomenon was captured. Additionally, the effect of velocity slip on the performance characteristics of GLJBs was studied. It was found that with the increase of eccentricity ratio and rotor speed, or a decrease of clearance size, the load-carrying capacity and friction torque of GLJBs rise, and the occurrence of velocity slip weakens the static performances of GLJBs.

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1. Introduction

With the rapid development of rotating machinery, bearings as their support elements are receiving increasing attention. GLJBs adopt gas as the lubricating medium and provide load-carrying capacity by rotating the journal and driving the gas movement in the wedge gap to form the dynamic

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Nomenclature

C	Radial clearance (m)	u_θ, u_r	Velocity components of the analytical solution
C_0	Average clearance (m)	$u_\theta^{num}, u_r^{num}$	Velocity components of the numerical solution
c	Lattice velocity	W_x	Dimensionless load along the line of center
c_s	Lattice sound speed	W_y	Dimensionless load perpendicular to line of center
$\tilde{c}_i$	Contravariant velocity	W	Net dimensionless load-carrying capacity
C_1, C_2	Accommodation coefficient	w	Weight coefficient
e	Eccentricity (m)	x, y	Coordinates on physical plane
E_{L_2-norm}	L_2 -norm error	Λ	Bearing number
f	Particle distribution function	δt	Time step
$\mathbf{F}$	Forcing term	ε	eccentricity ratio
h_{max}	Maximum clearance (m)	l	CFL number
h_{min}	Minimum clearance (m)	φ	Attitude angle (°)
H	Dimensionless gas film thickness	κ_1, κ_2	Slip coefficient
h	Characteristic length (m)	λ	Mean free path of molecules (m)
Kn	Knudsen number	μ	Dynamic Viscosity (N·s/m ²)
$\mathbf{M}$	Transform matrix	θ	Circumference coordinate
Ma	Mach number	ρ	density
N	Number of lattices	σ_1, σ_2	Tangential momentum accommodation coefficient
O_j	Center of journal	τ_s	Relaxation time
O_b	Center of bearing casing	ω	Rotor speed (rad/s)
P	Dimensionless pressure	ξ, η	Coordinates on computational plane
P_a	Ambient pressure	$\Delta \xi_{up,i}$	Distance of streaming
R	Radius of the shaft (m)		
R_g	Gas constant		
$\mathbf{S}$	Diagonal relaxation matrix		
T	Temperature (K)		
T_j	Dimensionless friction torque		
t	time		
$\mathbf{u}$	Velocity vector		
$u_{slip,1}$	First-order slip velocity		
$u_{slip,2}$	Second-order slip velocity		
u_f	Local velocity adjacent to the wall		
u_w	Local velocity at the wall		
u_{ff}	local velocity closest to the node f		

Subscripts

α, β	Coordinate direction indices
i, j	Discrete velocity number

Superscripts

eq	Equilibrium
neq	Non-equilibrium
*	Post-collision state

pressure compression effect. GLJBs are widely applied in rotating machinery [1] such as high-speed machine tools, centrifuges, compressors, fans, and electronic computer memory devices because of the advantages of high operating speed, long working life, low friction resistance, and no pollution to the lubricated surface. Since the clearance of GLJBs is very narrow and usually at the micron level, the molecular mean free path (λ) cannot be neglected compared to the size of clearance [2–3]. On this occasion, the gas flow already has a rarefaction effect and the Knudsen number ($Kn = \lambda/h$, h is the characteristic length) is introduced to describe the degree of rarefied gas. When $Kn > 0.001$, the airflow is in the slip regime, and the discontinuity of velocity and temperature near the boundary should be considered.

The kinetic characteristics of gas bearings and the influence of velocity slip on their performance by applying the slip-modified Reynolds equation were investigated by numerous academics. The most commonly modified models are as follows: the first-order slip-modified Reynolds equation proposed by Burgdorfer [4], the second-order slip-modified Reynolds equation proposed by Hisa [5], the 1.5-order slip-corrected Reynolds equation proposed by Mitsuya [6], and the Fukui-

Kaneko (FK) model [7–8] derived from the linearized Boltzmann equation (BE). However, the first-order and second-order slip modified Reynolds equations will inaccurately evaluate the bearing capacity, and both will rapidly deviate from the exact solution under high Kn number conditions. The FK model has proven to be the most accurate and available, while it is highly dependent on the created flow database and consequently is complicated to use [9]. Therefore, further modifications were carried out based on the above four models. Hwang et al. [10] developed a higher order slip-modified Reynolds equation model based on three adjustable coefficients, corrected the adjustable parameters according to the linearized BE, and further relaxed the restriction on Kn number. Zhang et al. [11] introduced effective viscosity based on the first-order slip-corrected Reynolds equation, and it showed that the existing model fits better with the FK model. Li et al. [12] introduced the angular deviation and sparse effect based on the FK model to analyze the impact of eight dynamic characteristic coefficients on MEMS.

The aforementioned studies assume that the gas flow is continuous. However, when the clearance is very narrow or the temperature is relatively high, the Kn number will be greater

than 1. The airflow in microchannel is already in the transition region, which does not satisfy the continuity assumption and the classical Reynolds equation will no longer be applicable. Hence it needs to introduce numerical methods depending on microscopic gas kinetics theory, such as LBM, direct simulation Monte Carlo method, molecular dynamics, etc.

The LBM has been demonstrated to be a powerful tool for studying rarefied gases as a mesoscopic method [13]. It carries the advantages of simple programming, the ability to handle complex boundaries, and ease of parallel computing. Jhon's team [14–18] earlier proposed a simulation method for gas bearings based on the LBM, which can calculate both the velocity and pressure fields. Subsequently, they adopted the truncated power-law model and the Bird-Carreau model to connect the shear rate and relaxation time, established a 2D/3D model suitable for non-Newtonian extremely thin liquid films with ultra-high shear rates. The Langmuir slip boundary condition is also applied. Additionally, they extended the LBM model using spatially dependent relaxation times to predict multicomponent gas mixture air bearings and studied air-helium mixture air bearings, predicting the velocity, pressure, and flow rate well. Zhou et al. [19] used LBM to analyze the phenomena at moderate and high Kn numbers and successfully captured the Kn minimum, validating that the LBM can be employed in the transition flow region. Kucinski et al. [20] discussed the application of various wall boundary conditions and the MRT model in gas film lubrication. This model does not ignore the inertial force and introduces the bearing cavitation algorithm. Sajjadi et al. investigated the natural convection flow of Cu/water [21] and MWCNT–Fe₃O₄/water [22] nanofluids in porous cavities with sinusoidal temperature distribution using a new double MRT lattice Boltzmann method. The MRT model has shown superior performance in the field of complex flow problems such as particle deposition [23], magnetohydrodynamics, and porous media [24]. Nickaeen et al. [25] applied the bounce back boundary to study the macroscopic gas bearing, while applying the diffusion bounce back (DBB) boundary to study the microscopic air bearing. It indicated that the gas film pressure decreased with the Kn increased. Zhang et al. [26] studied high speed and temperature gas bearings of wedge shape employing the finite difference LBM (FDLBM) and compared it with the macroscopic method. They found that the gas film pressure predicted by FDLBM was smaller than that predicted by the macroscopic method. Jiao et al. [27] established three infinitely wide wedge models with the shape of linear, parabolic, and harmonic, and studied the streamlines and pressure distributions of different wedges using the LBM. The results showed that the harmonic wedge model had the larger peak pressure and load-carrying capacity. Unfortunately, all these studies are aimed at wedge-shaped gas bearings, and no research has been conducted on GLJBs with curved variable cross-section microchannels.

For microchannels with complex geometries, the treatment of meshes is particularly important, and currently body-fitted grids are conventional and appropriate. He & Doolen [28] first implemented the interpolation supplemented LBM (ISLBM) on a nonuniform grid. Compared with the standard LBM, there is an extra interpolation process, but the number of grids is reduced and the time spent does not increase significantly. Subsequently, Imamura [29–30] proposed the GILBM on the above basis, which was successfully adopted for simulating

the flow around cylinders and airfoils. More complex flow domains such as a rectangle channel in a 180 bend [31], concentric and eccentric cylindrical annuluses [32], and cylindrical Couette flow [33] have also been studied based on the GILBM. Immersed boundary lattice Boltzmann method (IB-LBM) [34–35] is another effective method to deal with the flow involving curved walls. It is an Eulerian-Lagrangian model that has achieved great success in complex fluid-structure interaction problems with rigid and flexible walls [36–37]. IB-LBM will be a very potential research tool in the study of dynamic characteristics such as deformation of flexible foil and journal disturbance of GLJBs.

In this paper, the body-fitted grid is adopted to divide the flow domain of curved variable cross-section microchannel and then solved by GILBM. For the treatment of slip boundary, the first-order and second-order slip boundary condition is implemented when the airflow is in the slip region and transition region separately. Due to the presence of variable cross-section, the gas film thickness equation was used to modify Kn and the functional relationship between viscosity and density was also introduced. Based on the above model, the effect of velocity slip on the performance of GLJBs with different eccentricity ratios, rotor speeds, and clearance sizes were investigated.

2. Methodology

2.1. The physical model

The schematic diagram of a GLJB is delineated in Fig. 1. The radii of shaft and housing are R and $R + C_0$, respectively, where C_0 is the clearance size under concentric, $h_{\max}$ and $h_{\min}$ are the maximum and minimum clearances, respectively. The journal rotates clockwise with speed ω , the eccentricity ratio is $\varepsilon = e/C_0$, where e is the eccentricity. The angle between load W and the O_jO_h line is defined as the attitude angle φ . The dimensionless gas film thickness equation is

$$H = 1 + \varepsilon \cos \theta \quad (1)$$

Assuming that the Kn number is Kn_0 in the case of concentricity. When the eccentricity is ε , the Kn number at different circumferential angles θ is

$$Kn(\theta) = Kn_0 / (1 + \varepsilon \cos \theta) \quad (2)$$

After calculating the gas film dimensionless pressure (P), the dimensionless load-carrying capacity W of the GLJB will be obtained as

$$W_x = \int_0^{2\pi} (P - 1) \cos \theta d\theta \quad (3)$$

$$W_y = \int_0^{2\pi} (P - 1) \sin \theta d\theta \quad (4)$$

$$W = \sqrt{W_x^2 + W_y^2} \quad (5)$$

Where W_x , W_y are the non-dimensional loads along the centerline and perpendicular to the centerline, respectively.

The positive direction of friction torque is defined as the same direction as the journal rotation. The circumferential dimensionless friction torque on the surface of a journal is

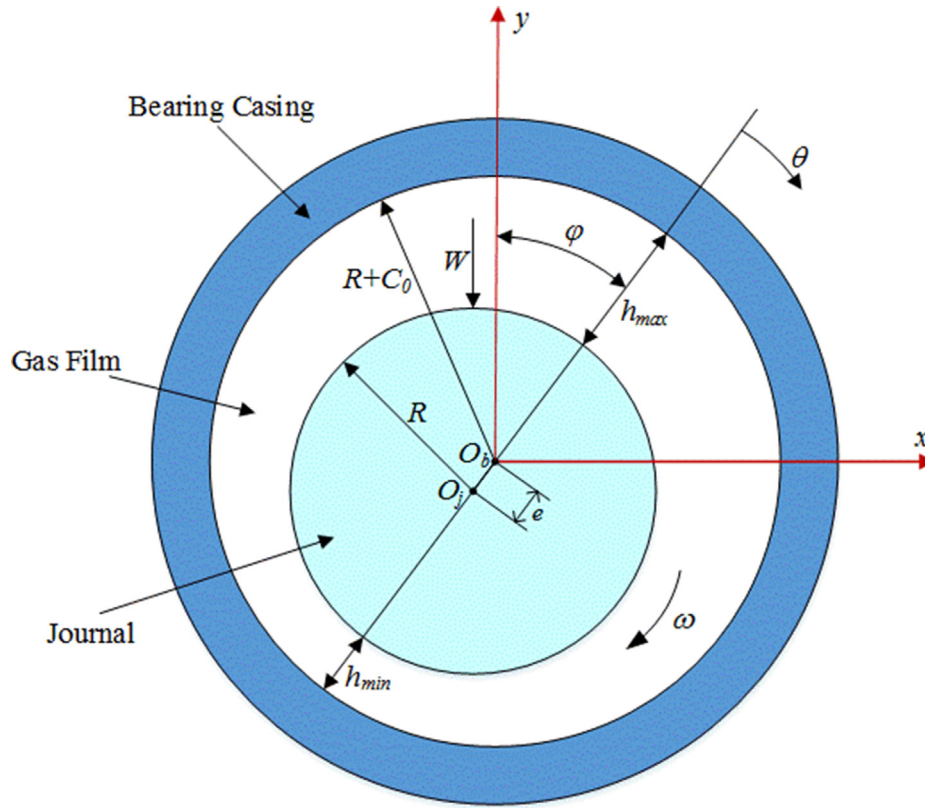


Fig. 1 Diagrammatic sketch of a GLJB.

$$T_j = \int_0^{2\pi} \left(\frac{H}{2} \frac{\partial P}{\partial \theta} + \frac{\Lambda}{6} \frac{1}{H} \right) d\theta \quad (6)$$

Where $\Lambda = \frac{6\mu\omega}{P_a} \left(\frac{R}{C_0} \right)^2$ means the bearing number, P_a is the ambient pressure, μ is the viscosity of the lubricant.

The structural parameters and operating conditions of GLJBs are provided in Table 1.

2.2. The multiple-relaxation-time LBM

The LBM evolved from lattice gas automata (LGA) and is a particular discrete format of the simplified Boltzmann equation. One of the most widely used models for LBM is the single relaxation time (SRT) model. However, recent studies have shown that the SRT model has problems with the simulation boundaries for microflows and also contains some nonphysical numerical artifacts [38]. To solve these problems, the MRT model [39] is adopted, with the following evolution function,

$$f_i(x + c_i \delta t, t + \delta t) - f_i(x, t) = -\sum_j (M^{-1} S M)_{ij} [f_j - f_j^{(eq)}] + \delta t F_i \quad (7)$$

Where $f_i(x, t)$ is the distribution function of the particle in the orientation of discrete velocity c_i , at position x and time t , δt is the time step, $i, j = 0, 1, \dots, b$, b is the total number of discrete velocities, F_i is the discrete external force term, S is a diagonal relaxation matrix, M is a transform matrix, $f_j^{(eq)}$ is the equilibrium distribution function, which is related to fluid density ρ , velocity u , temperature T , and the specific formula is

$$f_j^{eq} = w_j \rho \left[1 + \frac{c_j \cdot u}{c_s^2} + \frac{(c_j \cdot u)^2}{2c_s^4} - \frac{u^2}{2c_s^2} \right] \quad (8)$$

Where w_j is a weighting coefficient that depends on the model, $c_s = \sqrt{R_g T}$ is the lattice sound velocity (R_g is the gas constant), which is usually equal to $c/\sqrt{3}$ for isothermal flow, c is the lattice velocity, $c = \delta x/\delta t$, and δx is the lattice space step size. Here, the Mach number ($Ma = \omega R/c_s$, ωR is the linear velocity of the shaft) is introduced to facilitate the introduction of bearing speed.

In this paper, the two-dimensional case is considered and the D2Q9 model is selected, with the nine discrete velocity directions.

$$c_i = \begin{cases} (0, 0), & i = 0 \\ [\cos(i-1)\pi/2, \sin(i-1)\pi/2], & i = 1, 2, 3, 4 \\ \sqrt{2}[\cos(2i-1)\pi/4, \sin(2i-1)\pi/4], & i = 5, 6, 7, 8 \end{cases} \quad (9)$$

Table 1 GLJBs structural and operating parameters.

Parameter	Value
Radius of the shaft (R)	8 mm
Radial clearance (C)	10–75 μm
Speed of the shaft (ω)	30–110 krpm
Eccentricity ratio (ϵ)	0.2–0.8
Lubricant	Helium
Knudsen number (Kn)	0.027–0.324

The weight coefficients in Eq. (8) take the value of $w_0 = 4/9$, $w_1 = 1/9$, $w_5 = 1/36$, the diagonal relaxation matrix $\mathbf{S} = \text{diag}(\tau_p, \tau_e, \tau_e, \tau_d, \tau_d, \tau_q, \tau_q, \tau_s, \tau_s)^{(-1)}$, the relaxation times are set as $\tau_p = \tau_d = 1.0$, $\tau_e = 1.1$, $\tau_e = \tau_q = 1.2$, the transform matrix $\mathbf{M}$ can be found in the reference [39].

The macroscopic density ρ and velocity $\mathbf{u}$ of the fluid is obtained from the moments of the distribution function f_i ,

$$\rho = \sum_{i=0}^8 f_i, \quad \rho \mathbf{u} = \sum_{i=0}^8 c_i f_i \quad (10)$$

The solution of LBE is generally separated into two parts: collision and streaming,

$$\text{Collision: } f_i^*(x, t) = f_i(x, t) - \sum_j (\mathbf{M}^{-1} \mathbf{S} \mathbf{M})_{ij} (f_j - f_j^{eq}) \quad (11)$$

$$\text{Streaming: } f_i(x + c\delta t, t + \delta t) = f_i^*(x, t) \quad (12)$$

The above LBM model is applicable to continuous flow, and to enable the simulation of microflow, two important issues need to be addressed, one is the need to correlate the relaxation time τ_s with the Kn [40],

$$\tau_s = \frac{1}{2} + \sqrt{\frac{6}{\pi}} N Kn \quad (13)$$

Where N is the number of lattices along the feature-length direction, to associate the relaxation time with density, we have: $\tau_s' = 0.5 + (\tau_s - 0.5)/\rho$. The second problem is dealing with the slip boundary, which will be discussed later.

2.3. GILBM

In this section, the MRT model based on a uniform Cartesian coordinate system will be extended to a non-uniform curvilinear coordinate system. The governing equation of the physical plane $\mathbf{x} = (x_1, x_2) = (x, y)$ must be converted to the computational plane $\boldsymbol{\xi} = (\xi_1, \xi_2) = (\xi, \eta)$.

The conversion of the collision term to the generalized coordinate system is elementary, just replace $\mathbf{x}$ in Eq. (10) with $\boldsymbol{\xi}$ and get the following form

$$f_i^*(\boldsymbol{\xi}, t) = f_i(\boldsymbol{\xi}, t) - \sum_j (\mathbf{M}^{-1} \mathbf{S} \mathbf{M})_{ij} (f_j - f_j^{eq}) \quad (14)$$

After performing the collision step, the streaming step will be implemented, while the streaming step is not as simple as collision. As shown in Fig. 2, the distribution of $\mathbf{c}_1 = (1, 0)$ discrete velocity vectors in the physical and computational planes respectively [30]. It can be found that the $\mathbf{c}_1$ is consistent in the physical domain, while it varies in the computational domain with the change of grid nodes. Therefore, the contravariant velocity $\tilde{\mathbf{c}}_{i,\alpha}$ is established in the computational domain and is defined as

$$\tilde{\mathbf{c}}_{i,\alpha} = c_{i,\beta} \frac{\partial \xi_\alpha}{\partial x_\beta} \quad (15)$$

Then, to obtain the distance $\Delta \xi_{up,i}$ of particle streaming, the contravariant velocity is integrated over the time step δt ,

$$\Delta \xi_{up,i} = \int_0^{\Delta t} d\xi_i = \int_0^{\Delta t} \tilde{\mathbf{c}}_i dt \quad (16)$$

Subsequently, the streaming term in the computational plane becomes

$$f_i(\boldsymbol{\xi}, t + \delta t) = f_i^*(\boldsymbol{\xi} - \Delta \xi_{up,i}, t) \quad (17)$$

In Eq. (16), the positions of $f_i^*(\boldsymbol{\xi} - \Delta \xi_{up,i}, t)$ are usually not on the lattice nodes in the computational plane. Hence, they cannot be captured directly and an interpolation process is necessary. For the inner nodes, the second-order upwind interpolation method is applied to ensure the calculation precision. At the lattice points of adjacent boundaries, the second-order central difference interpolation method is applied since it is limited by the number of lattice nodes, and these corresponding interpolation functions can be found in Ref [30].

In GILBM, to keep the numerical stability, the determination of time step must satisfy the Courante-Friedrichs-Lewy (CFL) condition [41]. Referring to the traditional treatment in computational fluid dynamics, the time step δt is calculated by

$$\delta t = \iota \min_{i,\alpha} \left| \frac{1}{\tilde{\mathbf{c}}_{i,\alpha} \xi} \right| \quad (18)$$

Where ι is the CFL number ($0 < \iota \leq 1$), and in this work $\iota = 1$. This operation ensures that after a particle stream from a node, it will not exceed the range of the adjacent grid node, and a stable interpolation is guaranteed.

2.4. Langmuir slip model

Inspired by Kim et al. [16], the Langmuir slip model [42] is introduced in this section. Without considering the surface diffusion effect, the Langmuir slip model is applied to express the first-order slip velocity $u_{slip,1}$ as

$$u_{slip,1} = (1 - \kappa_1)u_f + \kappa_1 u_w \quad (19)$$

Where κ_1 represents the slip coefficient in the Langmuir slip model, u_f represents the local velocity neighboring wall, u_w represents the local velocity at wall. Solving Eq. (19) in conjunction with Maxwell's model, κ_1 can be derived as

$$\kappa_1 = \frac{1}{1 - C_1 Kn N}$$

Where C_1 is a parameter dependent on gas-wall interaction, $C_1 = 1.1466$, and $N = 1/\delta x$ is the number of grids.

However, concerning the transition regime, the first-order slip boundary no longer satisfies the accuracy requirements, thus the second-order slip velocity $u_{slip,2}$ represented by the Langmuir model is recommended [43].

$$u_{slip,2} = \frac{1}{1 - \kappa_2} [(1 - \kappa_1 - 2\kappa_2)u_f + \kappa_1 u_w + \kappa_2 u_{ff}] \quad (21)$$

Where u_{ff} denotes the local velocity closest to the node f , as shown in Fig. 3, and κ_2 is the second slip coefficient for the Langmuir model. After a similar derivation above, it can be gained.

$$\kappa_2 = -C_2 (Kn N)^2 \kappa_1 \quad (22)$$

Where C_2 is another parameter dependent on gas-wall interaction, and set as 0.9757.

The distribution function can be divided into equilibrium and non-equilibrium components according to the NEE scheme [44].

$$f_i(\mathbf{x}_w, t) = f_i^{eq}(\mathbf{x}_w, t) + f_i^{neq}(\mathbf{x}_w, t) \quad (23)$$

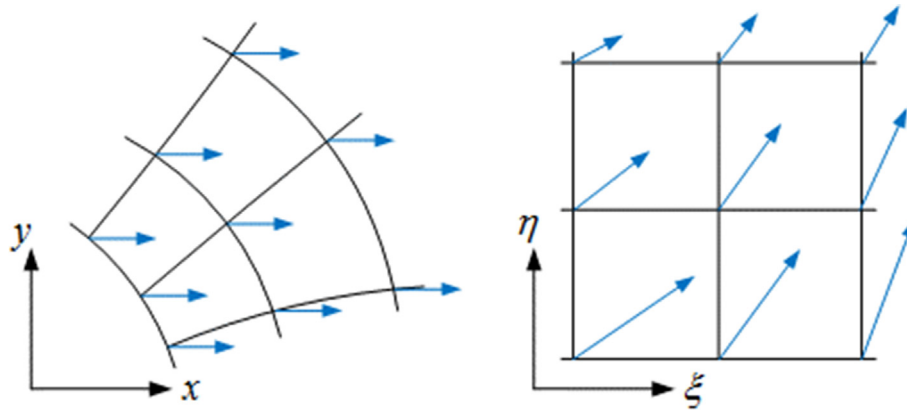


Fig. 2 $c_1 = (1, 0)$ discrete velocity vectors in the physical and computational planes.

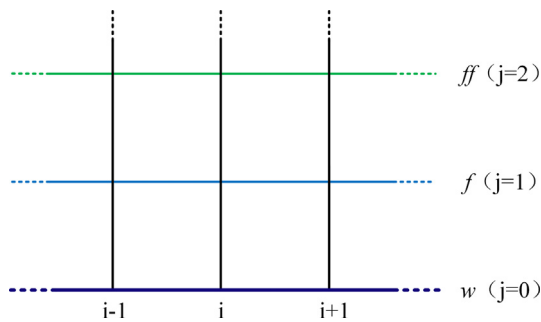


Fig. 3 Schematic of different node locations.

The Langmuir slip model was finally implemented in the NEE scheme, and the first-order slip boundary was obtained as follow

$$f_i(\mathbf{x}_w, t) = \kappa_1 [f_i^{eq}(\rho(\mathbf{x}_f, t), \mathbf{u}(\mathbf{x}_w, t)) - f_i^{eq}(\mathbf{x}_f, t)] + f_i(\mathbf{x}_f, t) \quad (24)$$

The second-order slip boundary condition can be derived as

$$f_i(\mathbf{x}_w, t) = \frac{1}{1 - \kappa_2} [\kappa_1 (f_i^{eq}(\rho(\mathbf{x}_f, t), \mathbf{u}(\mathbf{x}_w, t)) - f_i^{eq}(\mathbf{x}_f, t)) + (1 - 2\kappa_2)f_i(\mathbf{x}_f, t) + \kappa_2 f(\mathbf{x}_{ff}, t)] \quad (25)$$

It is worth noting that when $\kappa_2 = 0$, the above equation Eq. (25) degenerates to the first-order slip model, Eq. (24). Up to here, the slip boundary conditions are obtained that can be utilized in the slip flow and transition flow regimes.

2.5. Model verification

At first, we verify the precision of the model proposed in this work by simulating the microcylindrical Couette flow in the slip and transition region. The grid division is illustrated in Fig. 4, the number of grids in the circumferential and radial direction is $NX = 188$ and $NY = 20$, respectively. The inner cylinder of radius R_1 rotates clockwise with angular velocity ω and the outer cylinder of radius R_2 remains stationary, where $R_1 : R_2 = 3 : 5$. The analytic solution [45] of the dimensionless velocity distribution can be obtained by combining the Navier–Stokes (NS) equation with Maxwell’s diffusion model.

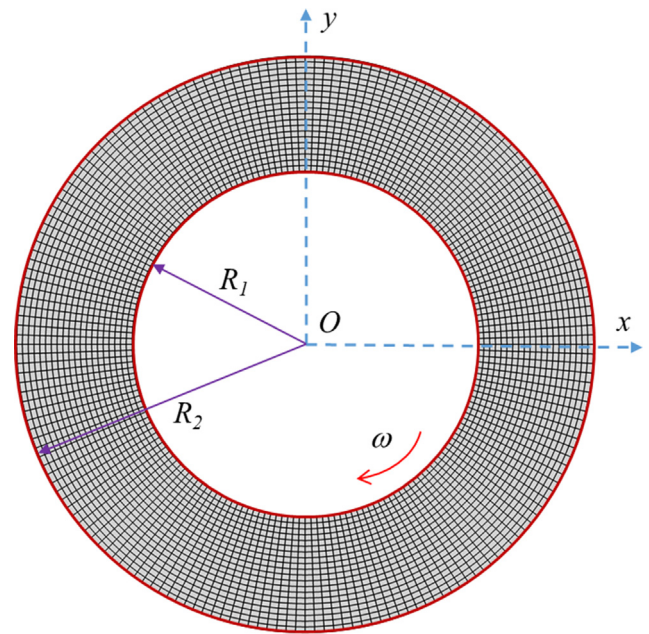


Fig. 4 Meshing of the microcylindrical Couette flow.

$$\frac{u_\theta}{\omega R_1} = \frac{1}{(A - B)R_1} \left(Ar - \frac{1}{r} \right), \quad u_r = 0 \quad (26)$$

With

$$A = \frac{1}{R_2^2} \left(1 - \frac{(2 - \sigma_2)}{\sigma_2} \frac{2\lambda}{R_2} \right) \quad (27)$$

$$B = \frac{1}{R_1^2} \left(1 + \frac{(2 - \sigma_1)}{\sigma_1} \frac{2\lambda}{R_1} \right) \quad (28)$$

Where u_θ , u_r are the tangential and radial velocity components, respectively, r is the distance from the circular center, σ_1 and σ_2 are the tangential momentum accommodation coefficient of the inner and outer cylindrical surfaces respectively, both taken as 1. The L_2 -norm error is utilized to evaluate the convergence of grids,

$$E_{L_2-norm} = \sqrt{\frac{\sum [(u_\theta - u_\theta^{num})^2 + (u_r - u_r^{num})^2]}{\sum (u_\theta^2 + u_r^2)}} \quad (29)$$

Where u_θ^{num} , u_r^{num} are the tangential and radial velocity components obtained from the present model, respectively.

The radial velocity profile for the microcylindrical Couette flow under different Kn numbers is depicted in Fig. 5(a). As can be seen, the results calculated by the present model agree well with the analytical solution in both slip flow and transition flow regions. The L_2 -norm error under different grid resolutions with Kn = 0.05 and Kn = 0.167 is shown in Fig. 5(b). The slope of the blue line is 1.98 and the slope of the red line is 2.03, indicating that the proposed model has second-order spatial accuracy.

3. Results and discussion

3.1. Grid independence verification

To eliminate the dependence of simulation results on the mesh numbers, a grid independence verification was carried out. The variation of the maximum gas film pressure with mesh numbers was examined for a GLJB with a radius of 8 mm, clearance of 25 μm , and speed of 50,000 rpm, as shown in Fig. 6. The physical domain was divided into five resolutions of 16157×7 , 22619×10 , 30159×15 , 45239×20 , and 60318×30 . It can be found that when the number of grids increased to a certain amount, the impact on the calculation results will be very slight. Considering the calculation accuracy and calculation time, the grid resolution of $30,159 \times 15$ was adopted.

3.2. Eccentricity ratio

In this part, the flow characteristics in the clearance of the GLJB were investigated when the eccentricity ratios were 0.2, 0.4, 0.6, and 0.8, respectively. The velocity contours and streamlines at the maximum gas film thickness for different eccentricity ratios were presented in Fig. 7(a)-(d). It can be found that as the eccentricity ratio increased, the proportion of low-velocity fluid expanded, implying that the blockage phenomenon gradually became obvious. When the eccentricity

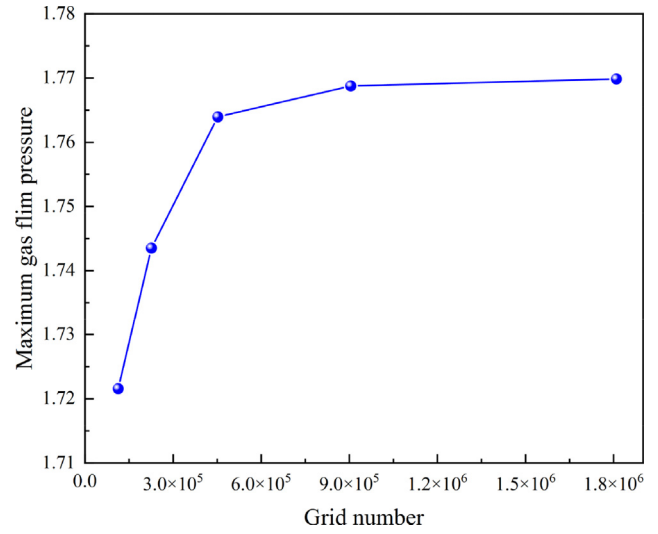


Fig. 6 Grid independence verification.

ratio enlarged to 0.8, the backflow phenomenon occurred and the airflow vortex formed.

The radial velocity profile at the maximum clearance with different eccentricity ratios was displayed in Fig. 8, including the slip and non-slip cases. It can be seen that regardless of whether slip existed, the radial velocity profile was approximately linear at a low eccentricity ratio, which was similar to the distribution pattern of Couette flow. However, as the eccentricity ratio increased, the velocity profile gradually bent downward, and finally even negative velocity values appeared, indicating that gas backflow occurred. More severe curvature of velocity profile indicated a stronger squeeze degree of air film. When velocity slip was considered, the velocity distribution curve lies below the no-slip case, implying that velocity slip led to a greater susceptibility to backflow and a greater degree of backflow.

The variation of dimensionless slip velocity on the journal surface with changed eccentricity ratios was depicted in Fig. 9. Interestingly, the variation pattern of slip velocity was similar to the variation pattern of the pressure profile given later. The maximum and minimum values of slip velocity appeared before

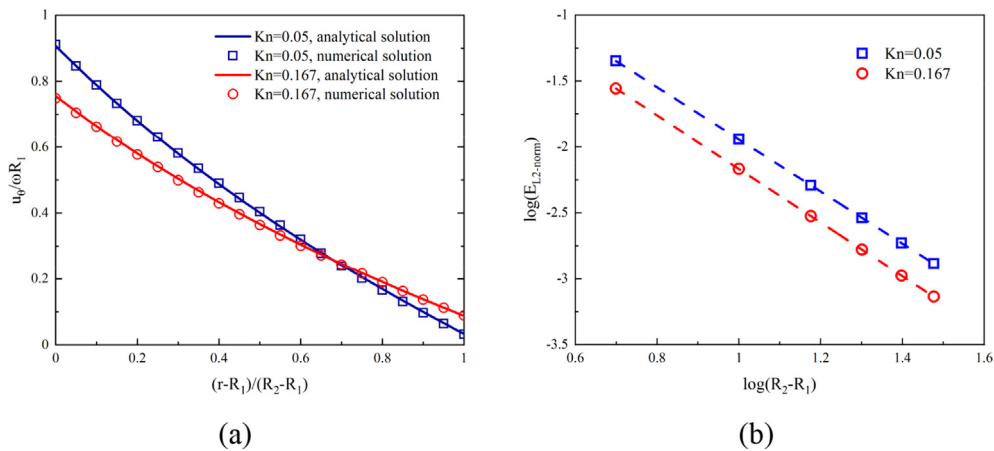


Fig. 5 (a) Radial velocity profile with different Kn numbers, (b) L_2 -norm error with different grid resolutions.

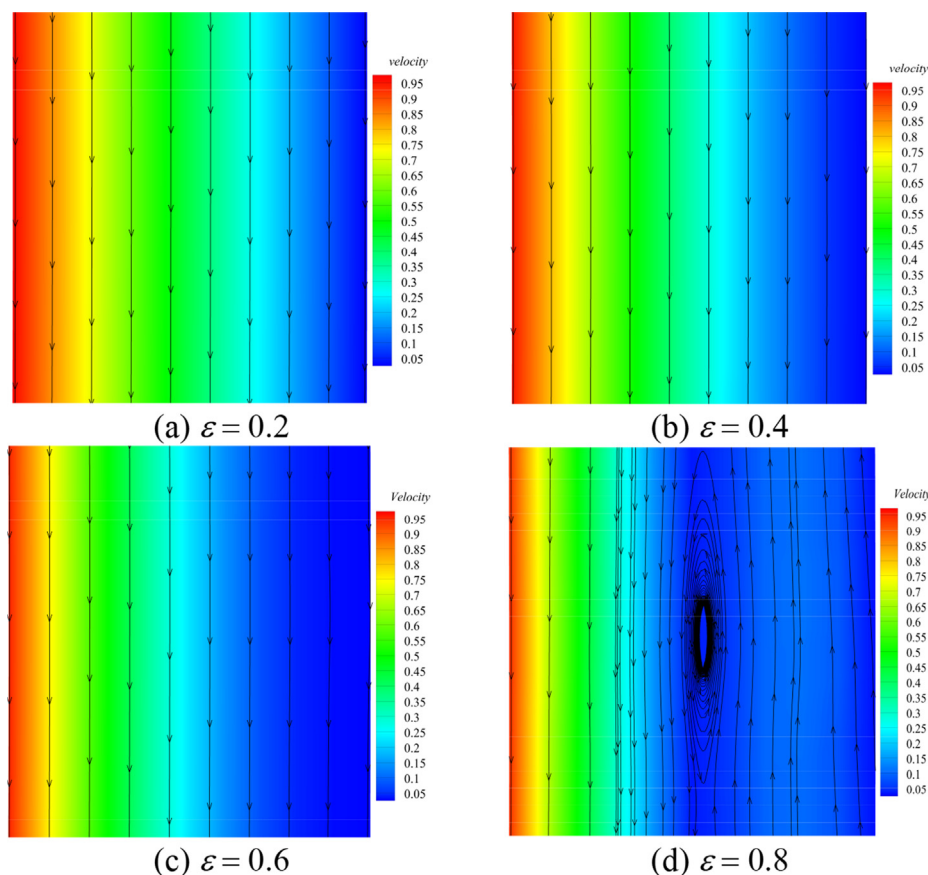


Fig. 7 Velocity contours and streamlines at the maximum clearance for various eccentricity ratios.

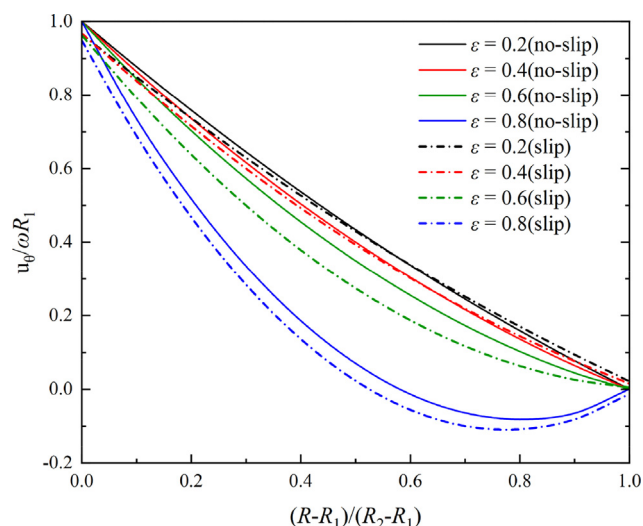


Fig. 8 Radial velocity profile in the maximum clearance for various eccentricity ratios.

and after reaching the minimum clearance, respectively. Because the bearing rotated clockwise, the gas flow was squeezed to the maximum extent before reaching the minimum gas film thickness, hindering the passage of gas flow, so the slip velocity is larger. The location after the minimum gas film thickness was the negative pressure gradient zone, with a force to promote fluid

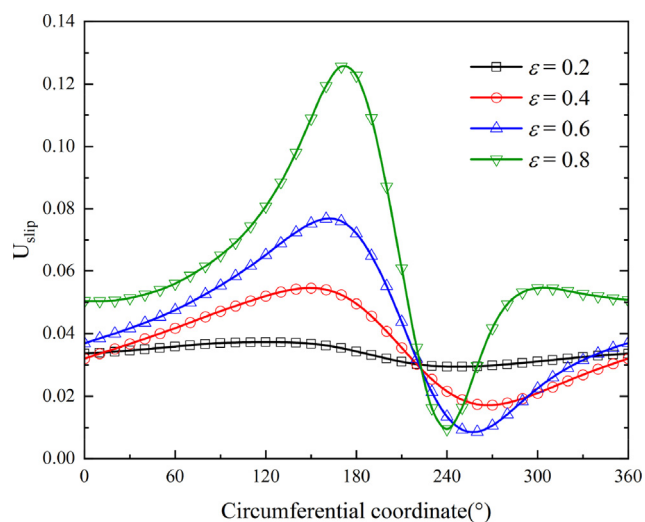


Fig. 9 Non-dimensional slip velocity on journal side for various eccentricity ratios.

movement, leading to the smaller slip velocity. In addition, with the increase of eccentricity, the maximum value of slip velocity increased, and the distinction in the circumferential distribution of slip velocity became progressively greater.

The dimensionless gas film pressure distribution for different eccentricity ratios in the no-slip and slip conditions was

depicted in Fig. 10(a). With the increase of eccentricity, the peak pressure of air film increased, and the shape of peaks and troughs became sharper, indicating a more concentrated pressure distribution. It can be found that due to the velocity slip, the gas film pressure below the non-slip case and the difference became progressively more distinct with the larger eccentricity ratio. In Fig. 10(b), the variation of dimensionless load and friction torque with eccentricity ratio in the cases of no-slip and slip was shown. As the eccentricity increased, the bearing capacity and friction torque rose and the acceleration of increase became faster. The occurrence of velocity slip reduced the bearing capacity and friction torque, and the reduction became more significant with the increase of eccentricity. It was because the slip weakened the dynamic compression effect and the frictional force at the wall.

3.3. Rotor speed

The radial velocity profile at the maximum gas film thickness with different rotor speeds was presented in Fig. 11. The velocity profile bent downward and had a negative value at low rotor speeds, indicating the occurrence of gas backflow. Gas flow blockage and backflow occurred due to the fact that the rate of airflow at the minimum gap was low. With the slow increase in speed, the velocity profile gradually bent upward, indicating that the influence of Poiseuille flow at the maximum clearance gradually enhanced. Besides, it seemed that the existence of velocity slip did not obviously influence the extent of backflow. As shown in Fig. 12, the distribution of dimensionless slip velocity on the bearing casing side with different rotor speeds was depicted. It can be found that the airflow velocity variation curve of the bearing clearance in the circumferential direction showed a similar distribution pattern with the slip velocity, proving that the slip velocity increased with airflow speed, explained by the increased velocity gradient. Moreover, the slip velocity on the bearing casing side also tended to increase with the rise of rotor speed, which also confirms the above discussion.

The distribution of dimensionless air film pressure with and without considering slip at various rotor speeds was presented in Fig. 13(a). As the rotor speed was raised from 30krpm to 110krpm, the dimensionless gas film peak pressure increased

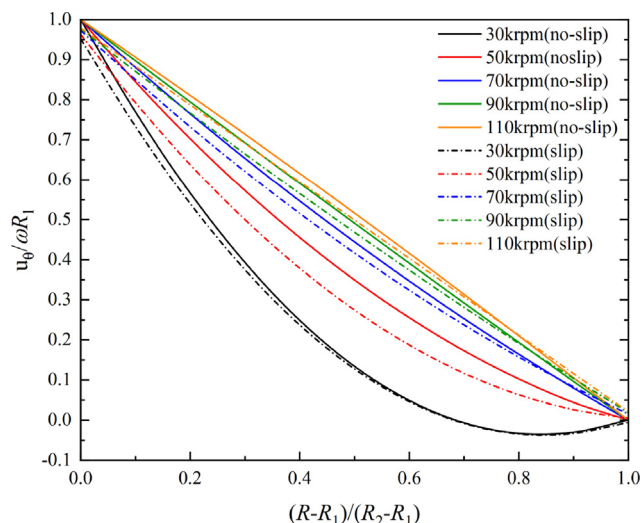


Fig. 11 Radial velocity profile at the maximum clearance with different rotor speeds.

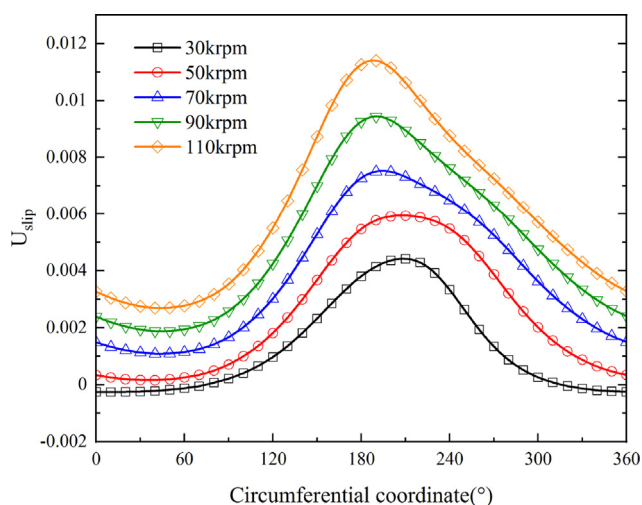
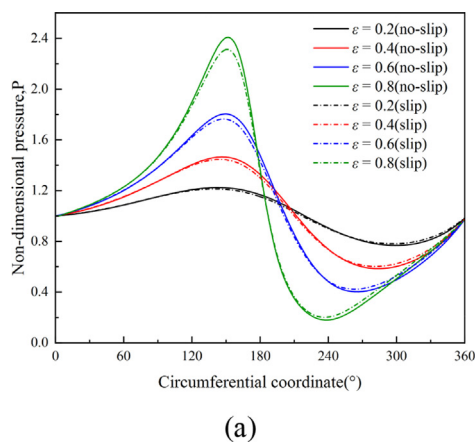
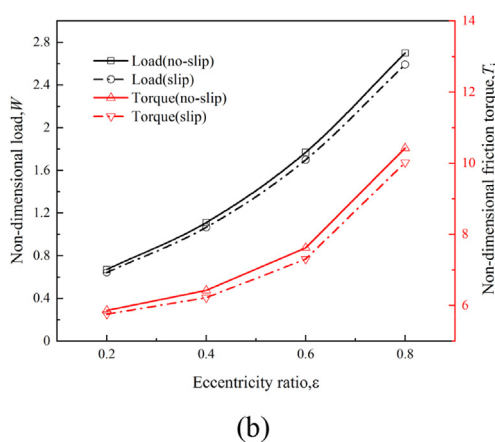


Fig. 12 Non-dimensional slip velocity on bearing casing side at different rotor speeds.



(a)



(b)

Fig. 10 (a) Non-dimensional air film pressure distribution for various eccentricity ratios, (b) Variation of dimensionless load and friction torque with eccentricity ratio.

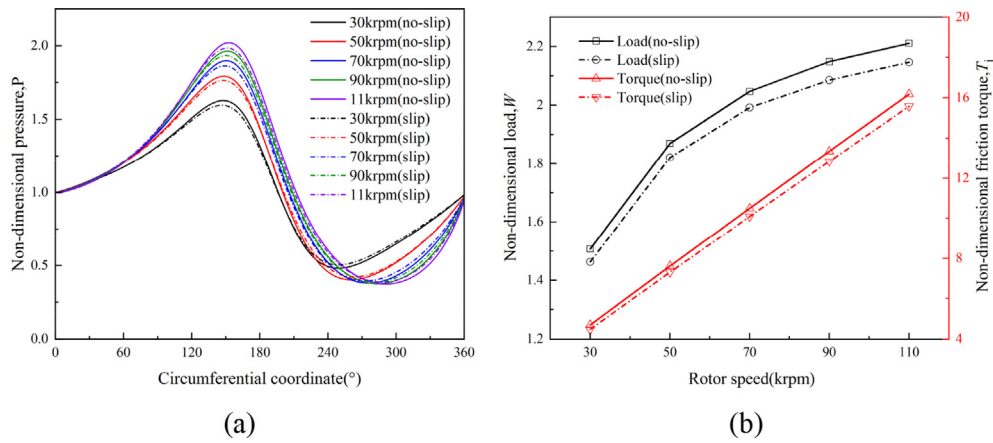


Fig. 13 (a) Non-dimensional gas film pressure distribution at different rotor speeds, (b) Variation of non-dimensional load and friction torque with rotor speed.

from fast to slow regardless of whether slip was considered. The location of dimensionless peak pressure remained essentially unchanged, but the location of dimensionless lowest pressure gradually migrated away from the minimum clearance. The alteration of dimensionless load-carrying capacity and friction torque with rotor speed was presented in Fig. 13 (b). As the rotor speed rose, the dimensionless load first rose rapidly, and then gradually tended to level off, indicating that blindly increasing the rotor speed did not necessarily bring great benefits. The appearance of velocity slip reduced the bearing capacity, and the reduction became more obvious with the raise of rotor speed. The reason here can be explained by the conclusion reached above that the slip effect became more severe with higher speed, weakening the compression effect to varying degrees. The friction torque enhanced almost linearly with increasing speed, that was $T \propto \omega$, which was consistent with the simple expression for the sliding bearing torque derived by Petroff. Since the occurrence of velocity slip will reduce the friction force, the friction force and torque were reduced even more with increasing speed.

3.4. Radial clearance

The effect of varying the bearing clearance size on the flow characteristics for a fixed eccentricity ratio and journal size was investigated. The effect of changing clearance size between 10 μm and 75 μm on the radial velocity profile at the maximum gas film thickness was depicted in Fig. 14(a). It can be found that with the widening of clearance size, the curvature of velocity profile gradually strengthened, the squeeze magnitude also enhanced, and the backflow phenomenon appeared. The reason was that compared with the small clearance size, the difference in size between the maximum clearance and minimum clearance in the case of a large clearance was greater, and more significant airflow obstruction was caused. The radial velocity profile at the minimum clearance for different clearance sizes when considering slip was shown in Fig. 14(b). At the clearance size of 10 μm , the velocity profile was linearly distributed, meaning that the Couette flow was absolutely dominant. As the clearance size increased, the velocity profile gradually curved upwards. When the gap was larger than 40 μm , the

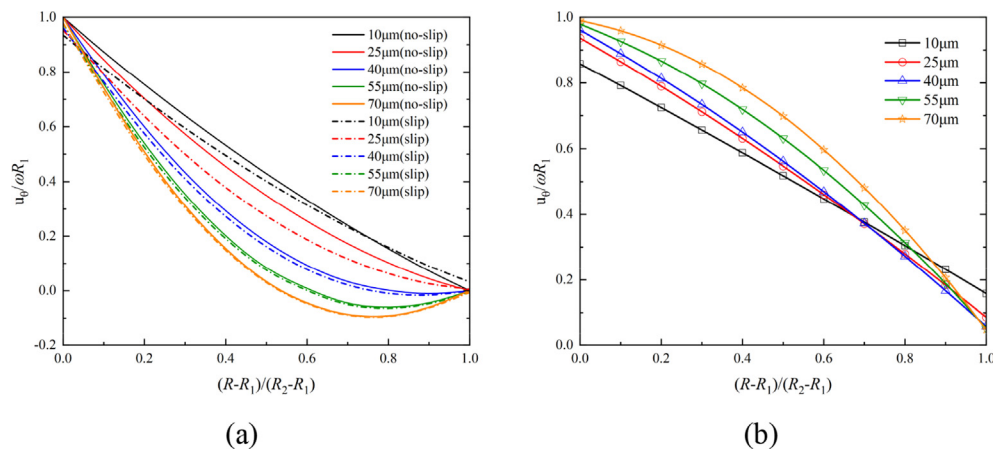


Fig. 14 Radial velocity profile in the maximum (a) and minimum (b) clearance under different clearance sizes.

curve had bent significantly, implying that airflow became a superimposed function of Couette flow and Poiseuille flow.

The variation of the non-dimensional pressure profile with the change of bearing clearance size was depicted in Fig. 15 (a). As the clearance size decreased, the gas film peak pressure raised first slowly and then quickly, and the position of the lowest pressure gradually deviated from the minimum clearance. The occurrence of boundary slip led to a reduction in the gas film peak pressure, and the smaller clearance led to more reduction in pressure. Because a small clearance implied a large Kn number, the gas rarefaction effect was more significant, and the velocity slip was larger, which weakened the dynamic pressure compression effect. It was worth mentioning that when the gap was 10 μm , the position of gas film peak pressure for the slip case was shifted compared with the no-slip case, further proving that the squeeze degree of air film was weakened by velocity slip. In Fig. 15(b), the non-dimensional load-carrying capacity and friction torque changed with clearance size. With the diminution of clearance size, both dimensionless load and friction torque tended to increase and the rate of increase kept rising. In particular, the friction torque rose sharply when the clearance was reduced from 25 μm to 10 μm , indicating that the frictional force becomes very large. The consideration of

the slip effect reduced the bearing capacity and friction torque, and the smaller clearance caused the larger Kn number, more rarefied gas, and weaker squeezing effect, while the enhanced velocity slip led to less friction.

3.5. Knudsen number

The Kn number of GLJBs clearance has always been a dimensionless number of great concern, because it may vary in a wide range for different bearing sizes and operating conditions, bringing inconvenience to the study. In the above research, there were many changing factors, and it was difficult to analyze the influence of Kn number on the airflow specifically. Therefore, based on the present model, the flow characteristics of airflow in the slip and transition regions with individually varying Kn numbers were investigated. In Fig. 16(a)-(b), the effects of changing Kn number on the dimensionless gas film pressure, dimensionless load-carrying capacity, and friction torque were displayed. It can be concluded that the increase of Kn number enhanced the gas rarefaction effect, weakened the dynamic pressure compression effect, and then reduced the peak pressure and bearing capacity. In addition, the

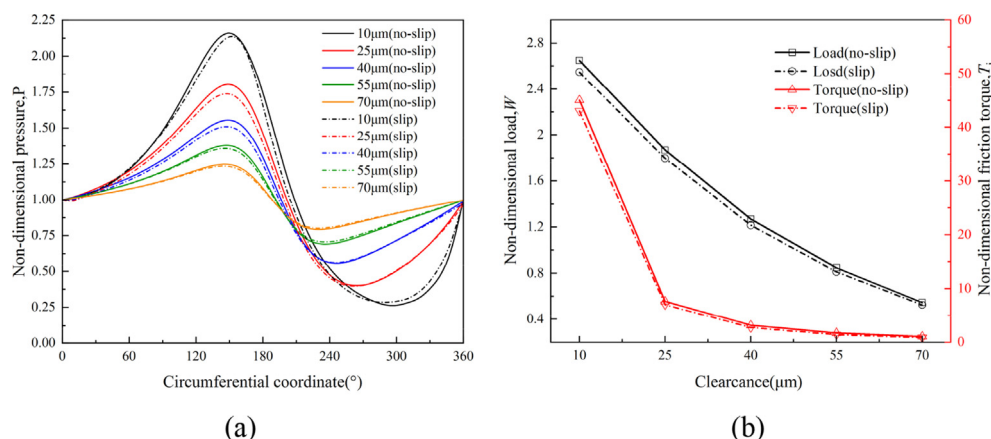


Fig. 15 (a) Non-dimensional gas film pressure distribution under different clearance sizes, (b) Variation of non-dimensional load and friction torque with clearance size.

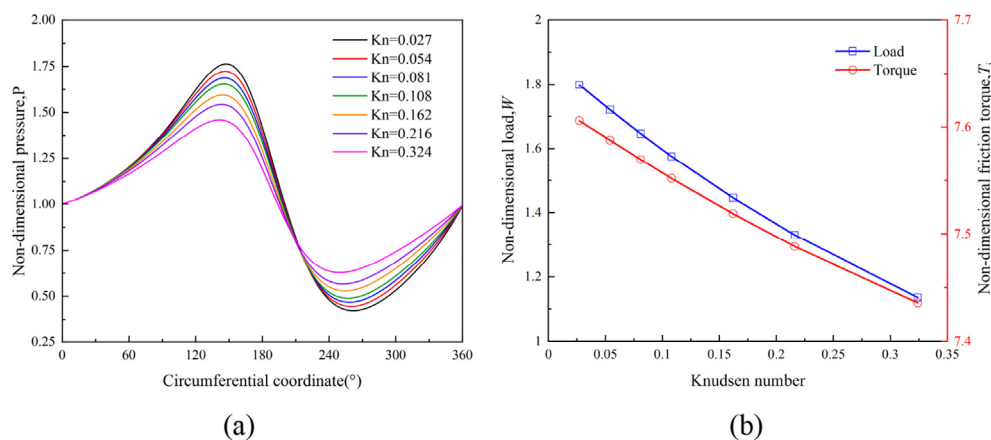


Fig. 16 (a) Non-dimensional gas film pressure distribution at different Kn numbers, (b) Variation of non-dimensional load and friction torque with Kn number.

enhancement of the slip effect reduced the frictional force and friction torque. Therefore, the rarefied gas will have a serious influence on the performance of the GLJB, and the study of airflow in bearing clearance at different Kn numbers has some guiding significance for the application of gas bearings in the high-altitude field.

4. Conclusions

In this work, a numerical model based on LBM to simulate the micro-flow in the clearance of GLJBs was established. The body-fitted grid was adopted to solve the complex geometric shape of the clearance, and the Langmuir slip model was introduced into the LB model based on the NEE scheme to obtain the first-order and second-order slip boundary conditions. Subsequently, the effects of velocity slip on the performance of GLJB under different bearing parameters and operating conditions were investigated, and the conclusions were as follows:

1. Compared with the traditional Reynolds equation method, the present method can be solved to gain the velocity profile in the bearing clearance and capture the gas backflow phenomenon at the maximum clearance under high eccentricity, low rotor speed, and large clearance size, with more microscopic details observed.
2. The present model can investigate the airflow of bearing clearance in the slip and transition regions, which makes up for the failure of the Reynolds equation in the transition flow region. The rarefaction effect of the lubricant weakened the performance of the GLJB with the addition of Kn number.
3. The occurrence of velocity slip at wall weakened the squeeze magnitude of air film, reducing the peak pressure and bearing capacity, while the slip effect decreased the frictional force and therefore the friction torque.
4. When the eccentricity ratio and rotor speed increased, or the clearance size decreased, the peak pressure of gas film, bearing capacity, and friction torque showed an increasing trend, with various increasing rates.

The model established in this paper can be applied to various complex-shaped microchannels and multi-scale simulation was realized, providing theoretical guidance for the study of gas bearings, Micro-electro-mechanical systems, and other micro-scale channels. However, in our study, the occurrence and degree of velocity slip were judged according to the size of Kn number. In fact, the timing and conditions of velocity slip in the actual operation of GLJBs are more complicated, this issue is worthy of further study in the future.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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